Vaibhav Nandkumar Kadam

Planning & Controls Engineer

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Education

Worcester Polytechnic Institute Worcester, MA
Master of Science in Robotics Engineering. 15th Aug 2022 - 1st May 2024
Ramrao Adik Institute of Technology, Mumbai University Mumbai, India
Bachelor of Engineering in Electronics Engineering. 04th Aug 2014 - 1st May 2018

Skills

- **Languages/Frameworks**: C++, Python, ROS2, Linux, Git, Docker, MATLAB/Simulink
- **Platforms**: NVIDIA Orin AGX, MABX, Vector CANape, ControlDesk
- **Simulation/Testing**: Gazebo, Carla, Real-World V&V, M-City

Experience

Magna International Troy, MI, USA
Research Engineer - Controls 23rd Sept 2024 - Present

- Developed object filtering and decision making modules for path planner with parked vehicle status classification, enabling safe obstacle avoidance maneuvers in urban delivery scenarios for **L4 Last Mile delivery robots.**
- Maintained and optimized NVIDIA Orin edge platforms for real-time planning and multi-sensor fusion, ensuring safety-critical latency requirements.
- Supported system integration and V&V, during field tests (M-City) and deployments in Toronto, ensuring the autonomy reliability .

Danfoss Power Solutions - Autonomy Minneapolis, MN, USA
Autonomy Systems Intern 15th May 2023 - 13th Sept 2024

- Designed decision-making architecture using behavior trees integrated with ROS2 autonomy stack (path following, obstacle detection).
- Built vehicle motion control interface with PACMoD ECU and Danfoss XM100, enabling safe drive-by-wire operation on Polaris Ranger AD prototype.
- Migrated autonomy modules to ROS2, validated in Gazebo simulations and deployed to Orin AGX hardware.

Peppermint Robots Pune, MH, India
Senior Robotics Engineer 19th July 2021 - 30th July 2022

- Led planning & controls of AMR autonomy stack with motion-primitive local planner + behavior trees, improving cluttered-indoor navigation.
- Delivered Nav2 local controller plugin for robust tracking and obstacle avoidance in warehouse environments.

NCETIS - IIT Bombay Mumbai, MH, India
Research Engineer 16th July 2018 - 28th Jun 2021

- Built Spherical Robot from concept to productization: multi-body dynamic modeling, nonlinear feedback linearization control, embedded deployment.
- Achieved 2 granted patents and industry adaptation of spherical robot design.

Major Projects

DigSafe [WPI, Industry Sponsorsed] Dec. 2022 - Dec 2023

- Developed an A* global planner leveraging HD Map data of power cable layouts for a differential-drive robot, designed a path-tracking control algorithm, and enhanced Electro-Magnetic (EM) sensor predictions using Kalman filtering for accurate buried-utility localization.

Valet Vehicle Parking using Hybrid Astar Aug. 2022 - Dec. 2022

- Implemented Hybrid-A* for planning Ackermann & Truck-Trailer considering non-holonomic kinematic constraints.

Short Range Path Planning using RL Aug. 2023 - Dec. 2023

- Implemented PPO algorithm for autonomous agent equipped with lidar to plan path in hospital environment over certian short distances avoiding obstacles.

For more projects please check **portfolio**

Patents

- **Earthquake Rescue and Relief Operations Robot**, Indian Patent 446857, **Granted August 23, 2023**
- **A Robot System with Upwards Facing Camera**, Indian Patent 462647, **Granted October 27, 2023.**
- **Methods for Scaling Spherical Robot**, Indian Patent 507996, **Granted February 07, 2024.**